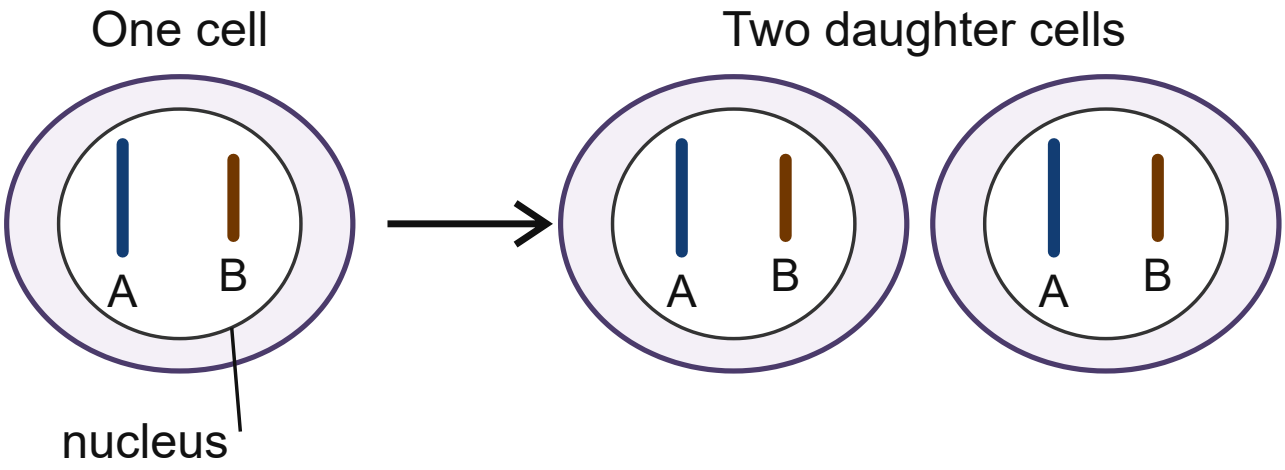
Reference A matching chromosome sets



Compare the full A and B set in both daughter cells. Each new cell has a matching set.

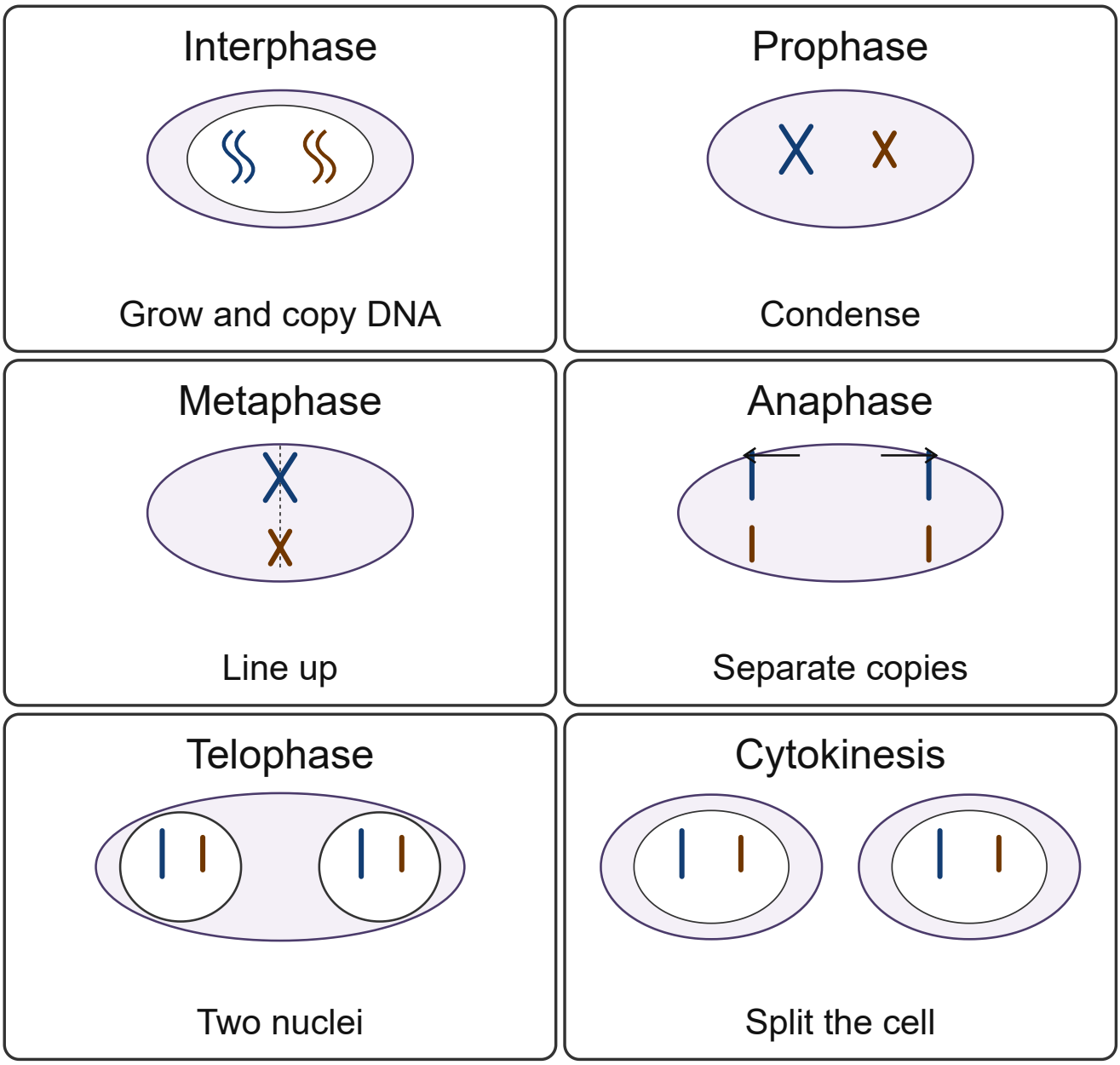
Reference only. Use labels as well as colours. The symbols are simplified chromosome types, not a full picture of real DNA.

Step	What changes
Interphase	The cell grows and copies its DNA.
Mitosis	Copied chromosomes separate into two nuclei.
Cytokinesis	The cell splits into two daughter cells.

In the number case, six chromosomes in the starting body cell means six in each daughter after ordinary mitosis and cell division. A fresh case starting with eight gives eight in each daughter.

Made by Matt

Reference B the six named stages



Interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Four middle phases belong to mitosis.

Reference only. Interphase comes before mitosis. Cytokinesis follows the formation of the two nuclei. The four named mitosis phases are prophase, metaphase, anaphase and telophase.

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Arrival and exit Independent

Use the labelled picture on Reference A for question 1. Respond in your usual mode.

1. Read the picture. How many daughter cells follow the final split?

2. Which labelled part of the first cell holds its chromosomes?

3. Compare the A and B labels in the two new cells. What do you notice?

4. What changed between the first and last pictures?

Exit ticket

1. Both A copies move to one end. Which phase first needs repairing?

2. Explain how the repair gives each new nucleus a matching set.

Pupil voice

What helped you most today, or what would you change next time?

Point, speak, sign, draw, write or choose. You may pass.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

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You do Independent

Use your book or this sheet. Use Reference A and B. One response per task.

1. Order interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Mark the four phases that belong to mitosis. [4]
Staff I ☐ S ☐ C ☐

---

2. Fault 1 goes into mitosis without copying DNA. Find the first wrong step and explain its repair. [3]
Staff I ☐ S ☐ C ☐

---

3. Fault 2 sends both A copies to one new nucleus. Find the first wrong step, repair it and explain the result. [3]
Staff I ☐ S ☐ C ☐

Feedback and response · Independent

Feedback codes

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

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